Zachary Anthony Cosenza
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Education

- PhD Chemical Engineering, University of California Davis (2018-2022)
- B.S. Chemical Engineering, University of Minnesota Twin Cities (2014-2018)

Work Experience

Data Science / Machine Learning Associate Consultant
ZS Associates, New York NY(Dec 2024 - Present)

- Developed and maintained production codebases for models predicting patient outcomes, providing daily recommendations on up to 10k national accounts, and supporting over 300k patients cross-brand.
- Built patient history transformer models to turn messy time-series data into embeddings, supporting patient risk segmentation efforts and interpretable feature creation.
- Lead efforts in feature engineering that improved the allocation of 13% of daily recommendations to a more accurate risk level, reducing burden on field teams.
- Developed agentic tools that leveraged unstructured and structured data and customized harnesses to enhance traditional score-based recommendations and provide real-time insights to field teams.
- Provided primary technical leadership for a team of data scientists/engineers while managing clients, third party contractors, and business constraints and timelines.

Applied Machine Learning Scientist Post-Doc (Dec 2022 - Dec 2024)
Lawrence Livermore National Laboratory, Livermore CA

- Trained deep multi-task and meta-learning models for protein drug binding prediction, resulting in 50% reduction in experimental cost via replacement of wet lab experiments with simulations.
- Fine-tuned biological language models to extract antibody features, resulting in 7-13% improvement in model performance on drug binding prediction tasks relative to physics-derived features.
- Contributed to research applying image segmentation for power-network mapping, and genetic algorithms for drug binding optimization under uncertainty.

Graduate Student Researcher (Aug 2018 - Dec 2022)
University of California Davis, Davis CA

- Developed multi-objective Bayesian optimization framework for experimental design under cost constraints and uncertainty, reducing required lab experiments by 70% vs traditional DOE methods.
- Published 5 first-author papers; invited speaker at ICML 2022 active learning workshop

Skills: Python (PyTorch, scikit-learn, pandas, PySpark), Bayesian and multi-objective optimization, genetic algorithms, multi-task and meta learning, language and transformer models, multi-GPU training DDP and SLURM, Kedro ETL, SQL, feature engineering, analytics and visualization.

Publications https://scholar.google.com/citations?user=O0UNOpcAAAAJ&hl=en

Personal Github (https://github.com/ZacharyCosenza): Topics include LLM pre-training via DDP, supervised fine-tuning (500M params, ~20B token data and model size), classification using semantic embeddings, RAG pipelines, knowledge graph creation, real-time prediction using open source APIs and orchestration via Argo.

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